

# Demonstration of Fonts and Alignment Filter for Pandoc

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This document demonstrates every feature provided by the `fonts-and-alignment` Lua filter for Pandoc. Each code block shows the exact Markdown syntax used to generate the rendered output that follows, ensuring consistent results across both LaTeX/PDF and HTML formats.<sup>1</sup>

The filter relies on Pandoc's `fenced_divs` and `bracketed_spans` extensions, which are typically enabled by default. In the unlikely event that these extensions are disabled, the Lua filter will display a terminal warning and pass over the corresponding divs and spans, causing them to appear as raw text in the rendered output.

## Bracketed Spans and Fenced Divs Invocations

The filter may be used with Pandoc's `bracketed_spans` and `fenced_divs` syntax extensions, as demonstrated below. Bracketed Spans are intended for styling smaller inline portions of text, while Fenced Divs are used for styling larger blocks of content. This follows Pandoc's standard syntax conventions. For more information on this syntax, please refer to the official Pandoc documentation: [Divs and Spans — Pandoc User's Guide](#).

```
[This font is extra extra large.]{.pfa-text-2xl}
```

This font is extra extra large.

```
::: {.pfa-text-2xl}
```

```
This font is extra extra large.
```

```
:::
```

---

<sup>1</sup>Requires including the companion `fonts-and-alignment.css` stylesheet in your HTML compilation.

This font is extra extra large.

## Font Sizing in Bracketed Spans

The filter provides nine predefined sizing hooks that allow text within Bracketed Spans to be scaled relative to the document's base font size.

| Size         | Syntax                     | Output |
|--------------|----------------------------|--------|
| tiny         | [Sample]{.pfa-text-3xs}    | Sample |
| scriptsize   | [Sample]{.pfa-text-2xs}    | Sample |
| footnotesize | [Sample]{.pfa-text-xs}     | Sample |
| small        | [Sample]{.pfa-text-s}      | Sample |
| normal       | [Sample]{.pfa-text-normal} | Sample |
| large        | [Sample]{.pfa-text-l}      | Sample |
| Large        | [Sample]{.pfa-text-xl}     | Sample |
| LARGE        | [Sample]{.pfa-text-2xl}    | Sample |
| huge         | [Sample]{.pfa-text-3xl}    | Sample |

## Font Sizing in Fenced Divs

The same sizing classes can also be applied to Fenced Divs, allowing an entire block of text to be rendered at a specific font size.

### Extra Small Font Size

```
::: {.pfa-text-xs}
```

This paragraph renders at the `_extra small_` font size.

```
:::
```

This paragraph renders at the *extra small* font size.

## Normal Font Size

```
::: {.pfa-text-normal}
```

This paragraph renders at the `_normal_` font size.

```
:::
```

This paragraph renders at the *normal* font size.

## Large Font Size

```
::: {.pfa-text-l}
```

This paragraph renders at the `_large_` font size.

```
:::
```

This paragraph renders at the *large* font size.

## Font Weights, Shapes, and Families in Bracketed Spans

The following typographic styles can be applied to Bracketed Spans to control weight, style, and typeface.

| Style      | Syntax                        | Output        |
|------------|-------------------------------|---------------|
| Bold       | [Sample]{.pfa-font-bold}      | <b>Sample</b> |
| Medium     | [Sample]{.pfa-font-medium}    | Sample        |
| Italic     | [Sample]{.pfa-font-italic}    | <i>Sample</i> |
| Slanted    | [Sample]{.pfa-font-slanted}   | <i>Sample</i> |
| Upright    | [Sample]{.pfa-font-upright}   | Sample        |
| Emphasis   | [Sample]{.pfa-font-emphasis}  | <i>Sample</i> |
| Serif      | [Sample]{.pfa-font-serif}     | Sample        |
| Sans-Serif | [Sample]{.pfa-font-sans}      | Sample        |
| Monospace  | [Sample]{.pfa-font-mono}      | Sample        |
| Small Caps | [Sample]{.pfa-font-smallcaps} | SAMPLE        |
| Normal     | [Sample]{.pfa-font-normal}    | Sample        |

## Font Weights, Shapes, and Families in Fenced Divs

These same typographic classes can also be applied to Fenced Divs, allowing an entire block of text to adopt a particular visual style.

### Bold Weight

```
::: {.pfa-font-bold}
This paragraph renders in bold type.
:::
```

**This paragraph renders in bold type.**

### Sans-Serif Family

```
::: {.pfa-font-sans}
This paragraph uses a sans-serif typeface.
:::
```

This paragraph uses a sans-serif typeface.

### Small Caps

```
::: {.pfa-font-smallcaps}
This paragraph is rendered in small caps.
:::
```

THIS PARAGRAPH IS RENDERED IN SMALL CAPS.

## Text Casing Transformations

Unlike purely visual styling in HTML/CSS—where `text-transform` only changes how text is rendered—these casing transformations operate at the [Pandoc Abstract Syntax Tree \(AST\)](#) level and modify the actual document content. This ensures that uppercase and lowercase conversions are preserved consistently across all output formats and remain intact when copied into other applications.

## Casing Transformations in Bracketed Spans

| Casing    | Syntax                   | Output |
|-----------|--------------------------|--------|
| Uppercase | [Sample]{.pfa-uppercase} | SAMPLE |
| Lowercase | [SAMPLE]{.pfa-lowercase} | sample |

## Casing Transformations in Fenced Divs

### Lowercase

```
::: {.pfa-lowercase}
THIS PARAGRAPH IS TRANSFORMED TO LOWERCASE.
:::
```

this paragraph is transformed to lowercase.

### Uppercase

```
::: {.pfa-uppercase}
this paragraph is transformed to uppercase.
:::
```

THIS PARAGRAPH IS TRANSFORMED TO UPPERCASE.

## Text Decorations for Bracketed Spans

Text decoration styles are implemented using the LaTeX `ulem` package, which is automatically included via the `header-includes` field whenever any of the corresponding classes are detected in the document. These styles are also mapped to equivalent CSS properties to ensure consistent rendering across LaTeX/PDF and HTML outputs. These decorations apply only to **Bracketed Spans**.

| Decoration       | Syntax                               | Output               |
|------------------|--------------------------------------|----------------------|
| Single Underline | [Sample]{.pfa-text-underline}        | <u>Sample</u>        |
| Double Underline | [Sample]{.pfa-text-underline-double} | <u><u>Sample</u></u> |
| Dashed Underline | [Sample]{.pfa-text-underline-dashed} | <u>Sample</u>        |

| Decoration       | Syntax                           | Output            |
|------------------|----------------------------------|-------------------|
| Dotted Underline | [Sample]{.pfa-text-uline-dotted} | <u>Sample</u>     |
| Wavy Underline   | [Sample]{.pfa-text-uline-wave}   | <u>Sample</u>     |
| Strikeout        | [Sample]{.pfa-text-strikeout}    | <del>Sample</del> |

**Note:** The `\xout{...}` marked-out text style has been removed from this version of the package due to lack of consistent cross-format support in HTML/CSS and limited usage in modern document workflows.

## Colors

This section describes how to apply color to text using the `pfa-font-color` attribute in both Bracketed Spans and Fenced Divs.

### Solid Colors

The `pfa-font-color` attribute supports [CSS3 named colors](#), full hexadecimal values, and three-digit shorthand hexadecimal values.

### Flexible Color Terminology

For standard solid colors, the filter automatically normalizes CSS3 color names. This means solid colors are completely case-insensitive and support various naming conventions. All examples below resolve perfectly to the CSS3 color `mediumvioletred` in both HTML and PDF formats:

| Naming Convention | Syntax                                      | Output |
|-------------------|---------------------------------------------|--------|
| Lowercase         | [Color]{pfa-font-color="mediumvioletred"}   | Color  |
| Title Case        | [Color]{pfa-font-color="Medium Violet Red"} | Color  |
| Kebab Case        | [Color]{pfa-font-color="medium-violet-red"} | Color  |
| Snake Case        | [Color]{pfa-font-color="medium_violet_red"} | Color  |
| Camel Case        | [Color]{pfa-font-color="mediumVioletRed"}   | Color  |
| Pascal Case       | [Color]{pfa-font-color="MediumVioletRed"}   | Color  |

| Naming Convention    | Syntax                                                   | Output |
|----------------------|----------------------------------------------------------|--------|
| Screaming Snake Case | <code>[Color]{pfa-font-color="MEDIUM_VIOLET_RED"}</code> | Color  |

## Applying Solid Colors

| Input Type                 | Syntax                                          | Output |
|----------------------------|-------------------------------------------------|--------|
| CSS3 Named                 | <code>[Sample]{pfa-font-color="crimson"}</code> | Sample |
| Hex Full                   | <code>[Sample]{pfa-font-color="#2E8B57"}</code> | Sample |
| Hex Shorthand <sup>2</sup> | <code>[Sample]{pfa-font-color="#666"}</code>    | Sample |

**Note:** While Pandoc’s default LaTeX template loads `x11names` (which includes numbered variants like `LightBlue3`), CSS and web browsers do not recognize these. To ensure your colors render perfectly across both PDF and HTML formats, you must stick strictly to the standard CSS3 Named Colors or Hexadecimal codes.

## Color Mixing

The filter natively supports LaTeX’s `xcolor` percentage mixing syntax, allowing you to tint, shade or mix colors on the fly. This translates perfectly into both PDF and HTML outputs.<sup>3</sup>

The mixing syntax uses the exclamation mark (!) to separate values:

| Mixing Type | Syntax Pattern                          | Description & Example                                                                                |
|-------------|-----------------------------------------|------------------------------------------------------------------------------------------------------|
| Tinting     | <code>BaseColor!Percentage</code>       | Blends with white.<br><code>Maroon!40</code> keeps 40% Maroon and 60% white.                         |
| Shading     | <code>BaseColor!Percentage!black</code> | Blends with black.<br><code>MediumVioletRed!80!black</code> keeps 80% MediumVioletRed and 20% black. |

<sup>2</sup>Three-digit shorthand expands by duplicating each single hexadecimal digit per RGB channel (e.g., `#666` expands to `#666666`).

<sup>3</sup>HTML output utilizes the native CSS `color-mix()` function to achieve the same blend.

| Mixing Type   | Syntax Pattern                | Description & Example                                                                            |
|---------------|-------------------------------|--------------------------------------------------------------------------------------------------|
| Two-Color Mix | BaseColor!Percentage!MixColor | Blends two specific colors.<br>RoyalBlue!50!ForestGreen<br>yields a 50/50 mix of both<br>colors. |

**[Warning] Strict Casing Rule:** Because mixed colors are passed directly to the LaTeX compiler, the flexible terminology rules do not apply here. You must use the exact casing expected by the LaTeX `xcolor` package, otherwise your PDF generation will break:

- **Base Colors:** The [19 core LaTeX colors](#) must be strictly **lowercase**.
- **Extended Web Colors:** The [CSS3 named colors](#) must be strictly **PascalCase**.

| Mixing Type      | Syntax                                                           | Output        |
|------------------|------------------------------------------------------------------|---------------|
| Tint (40% Base)  | <code>[Tinted]{pfa-font-color="Maroon!40"}</code>                | <b>Tinted</b> |
| Shade (80% Base) | <code>[Shaded]{pfa-font-color="MediumVioletRed!80!black"}</code> | <b>Shaded</b> |
| Mix (50/50)      | <code>[Mixed]{pfa-font-color="RoyalBlue!50!ForestGreen"}</code>  | <b>Mixed</b>  |

## Inheriting Colors in Fenced Divs

When applied to a Fenced Div, the `pfa-font-color` attribute defines the default text color for the entire block. All enclosed content inherits this color unless explicitly overridden by an inner Bracketed Span.

```
::: {pfa-font-color="DarkSlateGrey"}
The Fenced Div defines _DarkSlateGrey_ as the default text color for this
↪ block.
```

```
[This Bracketed Span overrides the inherited color to
↪ _tomato_.]{pfa-font-color="tomato"}
```

```
The remaining text continues using _DarkSlateGrey_ for the remainder of
↪ the Div.
:::
```



The Fenced Div defines *DarkSlateGrey* as the default text color for this block.  
**This Bracketed Span overrides the inherited color to *tomato*.**  
The remaining text continues using *DarkSlateGrey* for the remainder of the Div.

**Note:** The `pfa-font-color` utility is an attribute, not a class, and should not be prefixed with a period (`.`).

## Text Alignment within Fenced Divs

The `pfa-align-*` classes control text alignment within a Fenced Div while explicitly preserving hard line breaks (`\`). These utilities map directly to native LaTeX commands (`\raggedright`, `\centering`, and `\raggedleft`) in PDF output and equivalent CSS behaviors in HTML, making them ideal for formatting poetry, lyrics, or multi-line blocks.

### Left-aligned Text

```
::: {.pfa-align-left}
This block of text is _left-aligned_.
:::
```

This block of text is *left-aligned*.

### Center-aligned Text

```
::: {.pfa-align-center}
This block of text is _center-aligned_.
:::
```

This block of text is *center-aligned*.

### Right-aligned Text

```
::: {.pfa-align-right}
This block of text is _right-aligned_.
:::
```

This block of text is *right-aligned*.

## Line Break Preservation

```
::: {.pfa-align-center}
This block of text\
is _center-aligned_\
while preserving explicit line breaks.
:::
```

This block of text  
is *center-aligned*  
while preserving explicit line breaks.

## Block-Level Alignment

The `.pfa-block-*` utilities control the horizontal positioning of a Fenced Div as a whole, without affecting the internal text alignment of its contents. This allows the block itself to be positioned independently of how text is arranged within it.

### Left-Aligned Block

```
::: {.pfa-block-left}
The entire block is _left-aligned_.
:::
```

The entire block is *left-aligned*.

### Center-Aligned Block

```
::: {.pfa-block-center}
The entire block is _center-aligned_.
:::
```

The entire block is *center-aligned*.

### Right-Aligned Block

```
::: {.pfa-block-right}
The entire block is _right-aligned_.
:::
```

The entire block is *right-aligned*.

## Combining Multiple Classes

Multiple typographic utilities can be combined within the same element. Font, size, color, alignment, and decoration classes are designed to compose independently and can be applied together to both Bracketed Spans and Fenced Divs.

Class names are space-separated within {} following Pandoc attribute syntax. In addition to classes, key-value attributes such as `pfa-font-color` may be included alongside class definitions.

### Bracketed Span Composition

| Style                           | Syntax                                                                                      | Output        |
|---------------------------------|---------------------------------------------------------------------------------------------|---------------|
| Bold Sans-Serif                 | <code>[Sample]{.pfa-font-bold .pfa-font-sans .pfa-text-l pfa-font-color="red"}</code>       | <b>Sample</b> |
| Italic Monospace                | <code>[Sample]{.pfa-font-italic .pfa-font-mono .pfa-text-s}</code>                          | <i>Sample</i> |
| Small Caps, Underlined, Colored | <code>[Sample]{.pfa-font-smallcaps .pfa-text-underline pfa-font-color="forestgreen"}</code> | <u>SAMPLE</u> |

### Fenced Div Composition

Multiple utilities can also be applied to Fenced Divs to control alignment, typography, and color simultaneously. The resulting block inherits all specified styles while preserving Pandoc's standard attribute behavior.

```
::: {.pfa-align-center .pfa-font-sans .pfa-font-bold .pfa-text-l
↪ pfa-font-color="midnightblue"}
A centered, bold, sans-serif, large, midnight-blue Fenced Div
↪ demonstrating multiple combined utilities from the filter.
:::
```

**A centered, bold, sans-serif, large, midnight-blue Fenced Div demonstrating multiple combined utilities from the filter.**

## Legacy Aliases (Deprecated)

**Warning:** Legacy aliases are deprecated. They are retained strictly for backward compatibility and will be entirely removed in the next major release. New documents should use the `pfa-*` namespaces going forward.

### Font Sizing Aliases

| Legacy Class         | Modern Class                 |
|----------------------|------------------------------|
| <code>xsmall</code>  | <code>pfa-text-xs</code>     |
| <code>small</code>   | <code>pfa-text-s</code>      |
| <code>normal</code>  | <code>pfa-text-normal</code> |
| <code>large</code>   | <code>pfa-text-l</code>      |
| <code>xlarge</code>  | <code>pfa-text-xl</code>     |
| <code>xxlarge</code> | <code>pfa-text-2xl</code>    |
| <code>huge</code>    | <code>pfa-text-3xl</code>    |

### Font Weight, Shape, and Family Aliases

| Legacy Class            | Shorthand       | Modern Class                    |
|-------------------------|-----------------|---------------------------------|
| <code>bold</code>       | <code>bf</code> | <code>pfa-font-bold</code>      |
| <code>emphasis</code>   | <code>em</code> | <code>pfa-font-emphasis</code>  |
| <code>italic</code>     | <code>it</code> | <code>pfa-font-italic</code>    |
| <code>medium</code>     | <code>md</code> | <code>pfa-font-medium</code>    |
| <code>monospace</code>  | <code>tt</code> | <code>pfa-font-mono</code>      |
| <code>normalfont</code> | <code>nf</code> | <code>pfa-font-normal</code>    |
| <code>sans</code>       | <code>sf</code> | <code>pfa-font-sans</code>      |
| <code>serif</code>      | <code>rm</code> | <code>pfa-font-serif</code>     |
| <code>slanted</code>    | <code>sl</code> | <code>pfa-font-slanted</code>   |
| <code>smallcaps</code>  | <code>sc</code> | <code>pfa-font-smallcaps</code> |
| <code>upright</code>    | <code>up</code> | <code>pfa-font-upright</code>   |

## Alignment Aliases

| Legacy Class | Modern Class     |
|--------------|------------------|
| centering    | pfa-align-center |
| raggedleft   | pfa-align-right  |
| raggedright  | pfa-align-left   |

## Text Decoration Aliases

| Legacy Class | Shorthand | Modern Class          |
|--------------|-----------|-----------------------|
| uline        | u         | pfa-text-uline        |
| uuline       | uu        | pfa-text-uline-double |
| dashuline    | dau       | pfa-text-uline-dashed |
| dotuline     | dou       | pfa-text-uline-dotted |
| uwave        | uw        | pfa-text-uline-wave   |
| sout         | so        | pfa-text-strikeout    |

## Removed Classes

The following legacy classes were removed in version 2.0.0 and are no longer recognized by the filter. Existing documents should be updated to use the replacement classes shown below.

| Removed Class | Shorthand | Removed In | Replacement        |
|---------------|-----------|------------|--------------------|
| center        | —         | 2.0.0      | pfa-align-center   |
| flushleft     | —         | 2.0.0      | pfa-align-left     |
| flushright    | —         | 2.0.0      | pfa-align-right    |
| xout          | xo        | 2.0.0      | pfa-text-strikeout |